

SKILL : WEEK 6

Task -1: To Store Commands, Maintain History Buffer, Handle Capacity Limits, Retrieve Entries, Display History, Validate Consistency

Code:

```
1 01.c
2 #include <stdio.h>
3 #include <stdlib.h>
4 #include <string.h>
5
6 #define MAX_HISTORY 5
7 #define MAX_COMMAND 100
8 char history[MAX_HISTORY][MAX_COMMAND];
9 int history_count = 0;
10
11 /* Store a command in history */
12 void store_command(const char *command)
13 {
14     if (command == NULL || strlen(command) == 0)
15         return;
16
17     /* If buffer is full, remove the oldest command */
18     if (history_count == MAX_HISTORY) {
19         for (int i = 1; i < MAX_HISTORY; i++) {
20             strcpy(history[i - 1], history[i]);
21         }
22         history_count--;
23     }
24     strcpy(history[history_count], command);
25     history_count++;
26     printf("Command stored successfully.\n");
27 }
28
29 /* Retrieve a command by history number */
30 void retrieve_command(int number)
31 {
32     if (number < 1 || number > history_count) {
33         printf("Invalid history number.\n");
34         return;
35     }
36     printf("Command %d: %s\n", number, history[number - 1]);
37 }
38
39 /* Display all commands */
40 void display_history(void)
41 {
42     if (history_count == 0) {
43         printf("History is empty.\n");
44         return;
45     }
46     printf("\n----- Command History ----- \n");
47     for (int i = 0; i < history_count; i++) {
48         printf("%d %s\n", i + 1, history[i]);
49     }
50     printf("\n");
51 }
52
53 /* Validate history consistency */
54 void validate_history(void)
55 {
56     int valid = 1;
57     if (history_count < 0 || history_count > MAX_HISTORY) {
58         valid = 0;
59     }
60     for (int i = 0; i < history_count; i++) {
61
62     }
63 }
64
65 Get Help      WriteOut      Read File      Prev Pg      Cut Text      Cur Pos
66 Exit          Justify        Where is       Next Pg      UnCut Text   To Spell
```

Output:

```
1 screenkhilreddy@Stress-MacBook-Pro ~ % nano 01.c
2 screenkhilreddy@Stress-MacBook-Pro ~ % gcc 01.c -o 01
3 screenkhilreddy@Stress-MacBook-Pro ~ % ./01
4
5 ===== COMMAND HISTORY =====
6 1. Store Command
7 2. Retrieve Command
8 3. Display History
9 4. Validate Consistency
10 5. Exit
11 =====
12 Enter your choice: 1
13 Enter command: 1
14 Command stored successfully.
15 ===== COMMAND HISTORY =====
16 1. Store Command
17 2. Retrieve Command
18 3. Display History
19 4. Validate Consistency
20 5. Exit
21 =====
22 Enter your choice: 2
23 Enter history number: 2
24 Invalid history number.
25 ===== COMMAND HISTORY =====
26 1. Store Command
27 2. Retrieve Command
28 3. Display History
29 4. Validate Consistency
30 5. Exit
31 =====
32 Enter your choice: exit
33 Enter history number: Invalid history number.
34 ===== COMMAND HISTORY =====
35 1. Store Command
36 2. Retrieve Command
37 3. Display History
38 4. Validate Consistency
39 5. Exit
40 =====
41 Enter your choice: Enter history number: Invalid history number.
42 ===== COMMAND HISTORY =====
43 1. Store Command
44 2. Retrieve Command
45 3. Display History
46 4. Validate Consistency
47 5. Exit
48 =====
49 Enter your choice: Enter history number: Invalid history number.
50 ===== COMMAND HISTORY =====
51 1. Store Command
52 2. Retrieve Command
53 3. Display History
54 4. Validate Consistency
55 5. Exit
56 =====
57 Enter your choice: Enter history number: 2
58 Invalid history number.
59 ===== COMMAND HISTORY =====
60 1. Store Command
61 2. Retrieve Command
62 3. Display History
63 4. Validate Consistency
64 5. Exit
65 =====
66 Enter your choice:
67
```

Task -2: To Create Pipeline Structures, Store Commands, Maintain Execution Order, Connect Processes, Validate Pipeline Layout, Test Configurations

